

Single Trial detection toolbox for EPilepsy (Version 1.0)

— STEP1 User Manual (May 16, 2011)

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1 Introduction

This manual provides an overview of the Single Trial detection toolbox for EPilepsy (version 1.0: STEP1). STEP1 is an interactive Matlab toolbox for performing single-trial estimation of EEG (ERP/Epilepsy) parameters, both in the time domain and in the time-frequency domain. In the time domain, single-trial detection methods in STEP1 includes (1) Wavelet filtering (WF), (2) Multiple linear regression (MLR), (3) Multiple Linear Regression with Dispersion Term (MLRd), (4) the combination of WF and MLR (WF+MLR), (5) the combination of WF and MLRd (WF+MLRd). In the time-frequency domain, single-trial detection methods in STEP1 includes (1) Windowed Fourier transform (WFT) and (2) Matching Pursuit (MP).

2 Software requirements

STEP1 requires the MATLAB system to be installed. In addition, STEP1 is fully compatible with EEGLAB, which is an Open Source Matlab Toolbox for Electrophysiological Research. EEGLAB is required to be installed in the host computer.

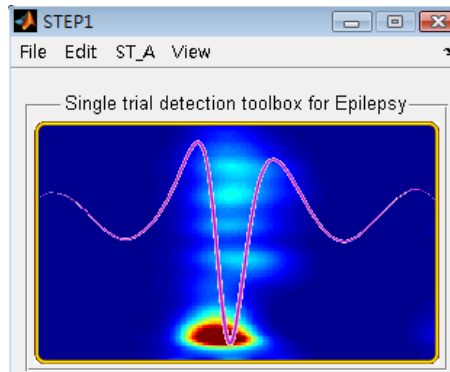
3 Installation, quick start guide & getting help

To install EEGLAB and STEP1:

1. Download (from [http://\\$\\$\\$\\$](http://$$$$)) and Unzip the file *eeglab7_2_9_20b.zip* and *STEP.zip*.
2. Start Matlab and list *eeglab7_2_9_20b* and *STEP* and all of their subdirectories in the Matlab search path (File→set path). Hit "save" in the pathtool window, which will make eeglab and STEP1 available for future Matlab sessions.

To start STEP1:

1. Start Matlab.
2. Type “**STEP1**” at the Matlab command line and hit enter. The main STEP1 window below should pop up, with its four menu headings: **File** **Edit** **ST_A** **View**.



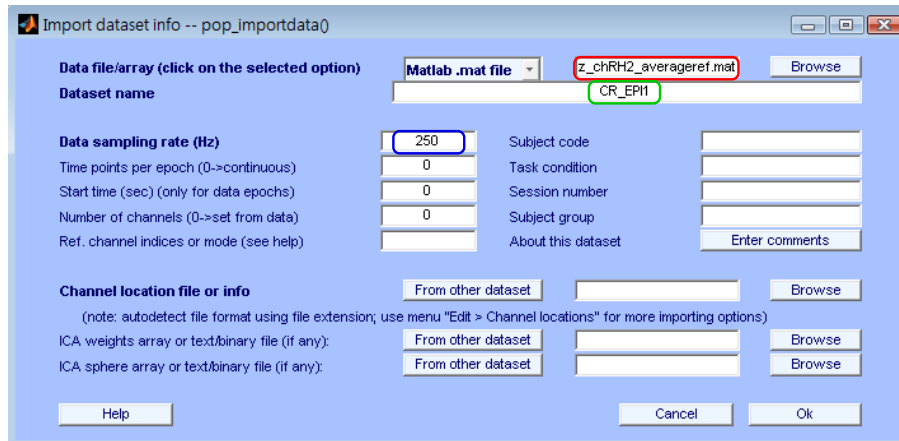
To get Help:

1. Search in this manual.
2. Refer to the following papers:
 - a) Hu, L., Mouraux, A., Hu, Y., Iannetti, G.D., 2010. A novel approach for enhancing the signal-to-noise ratio and detecting automatically event-related potentials (ERPs) in single trials. *Neuroimage* 50, 99-111.
 - b) Hu, L., Zhang, Z.G., Hung, Y.S., Luk, K.D., Iannetti, G.D., Hu, Y., 2011. Single-trial detection of somatosensory evoked potentials by probabilistic independent component analysis and wavelet filtering. *Clin Neurophysiol.*
 - c) Hu, L., Wise, R.G., Liang, M., Mouraux, A., Hu, Y., Iannetti, G.D. Taking into Account Latency, Amplitude and Morphology: Improved Estimation of Single-trial ERPs by wavelet filtering and multiple linear regression, *Journal of Neurophysiology*, in revision.
 - d) Zhang, Z., Luk, K.D., Hu, Y., 2010. Identification of detailed time-frequency components in somatosensory evoked potentials. *IEEE Trans Neural Syst Rehabil Eng* 18, 245-254.
 - e) Mayhew, S.D., Iannetti, G.D., Woolrich, M.W., Wise, R.G., 2006. Automated single-trial measurement of amplitude and latency of laser-evoked potentials (LEPs) using multiple linear regression. *Clin Neurophysiol* 117, 1331-1344.
 - f) Mouraux, A., Iannetti, G.D., 2008. Across-trial averaging of event-related EEG responses and beyond. *Magn Reson Imaging* 26, 1041-1054.
3. Email to Li Hu: hulitju@gmail.com.

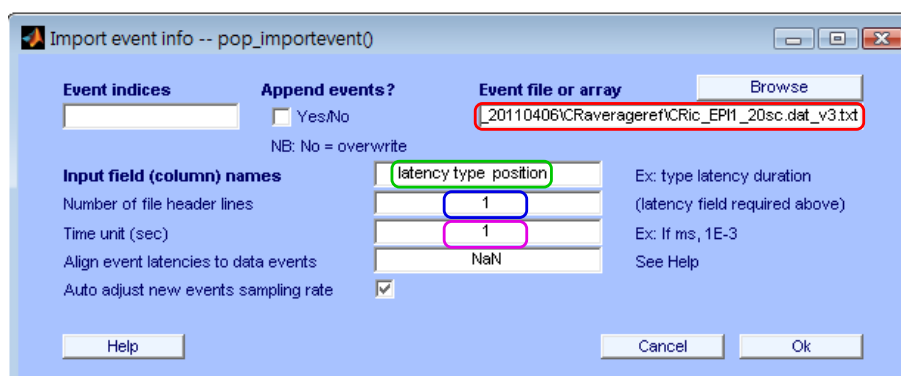
4 Importing/Exporting data into STEP1

To import data from Matlab array (2-D Matlab array should contain EEG data in which rows are channels and columns are data points):

1. Pop up Import dataset window by: File→Import EEG data
2. Specify the **data file**, the **Dataset name**, and the **Data sampling rate** accordingly (marked in red, green, and blue respectively), and click Ok.



3. Pop up Import event window by: File→Import Markers
4. Specify the **Event file**, the **Input field name**, the **Number of file header lines**, and the **Time unit (sec)** accordingly (marked in red, green, blue, and pink respectively), and click Ok.

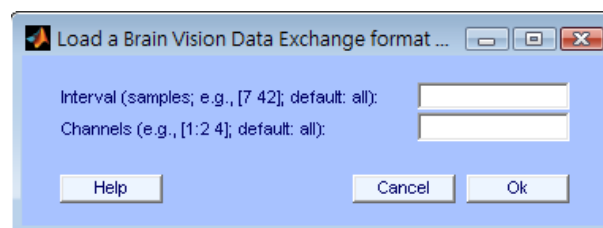


Note that the Marker file (.txt) should be arranged as the following way (three columns).

latency	type	position
19.48	mes_DH_DA	mes_DH_DA
19.908	mes_DA_DH	mes_DA_DH
20.464	mes_DH_DA_SB	mes_DH_DA_SB
21.388	mes_DA_DH	mes_DA_DH
22.476	mes_DH_DA	mes_DH_DA
22.912	mes_DH_DA	mes_DH_DA
23.468	mes_DH_DA_SB	mes_DH_DA_SB
24.42	mes_DA_DH	mes_DA_DH

To import data from Brain Vision Data Exchange format (.vhdr, .vmrk, .dat):

1. Pop up Import dataset window by: File→Import Brain Vision data, and select the .vhdr file.
2. Specify the **Time Interval** and the **Channels** (normally we accept the default setting and keep these fields in blank), and click Ok.



To load data in EEGLAB format:

1. Pop up load dataset window by: File→Load EEGLAB data
2. Specify the **file name and path**, and click Load.

To save data in EEGLAB format (saving frequently is recommended):

1. Pop up save dataset window by: File→Save→Save EEGLAB data
2. Specify the **file name and path**, and click Save.

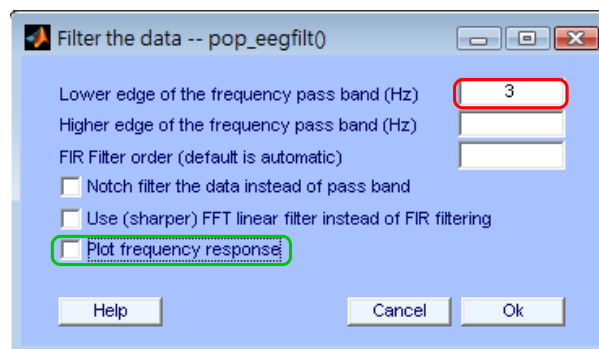
5 Pre-processing in STEP1

To filter the dataset:

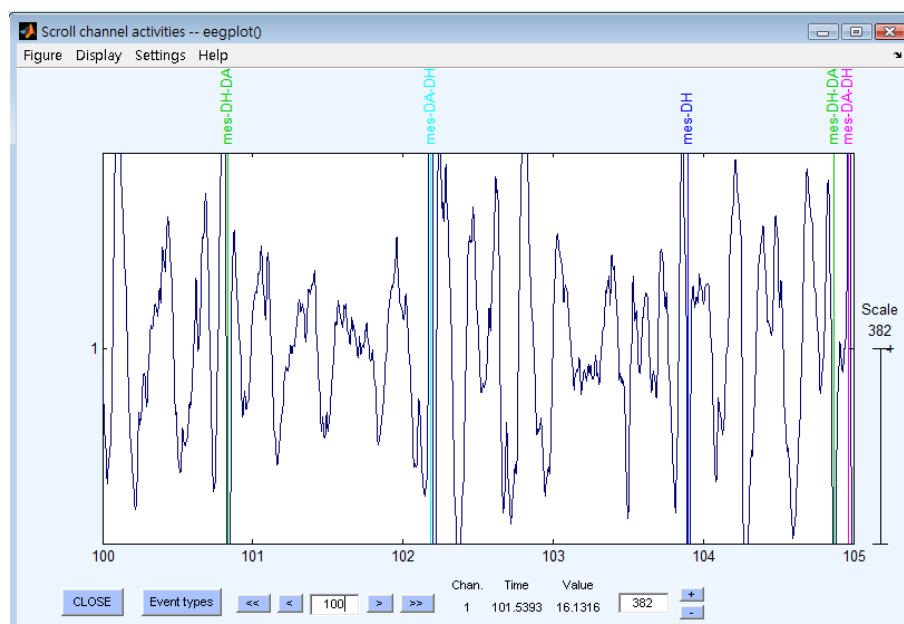
1. Pop up Filter the data window by: Edit→Filter

- Specify the **Lower edge of the frequency pass band** (*High pass filter*) and uncheck **Plot frequency response** (marked in red and green respectively), and click Ok.

Note that the **Higher edge of the frequency pass band** (*Low pass filter*) can also be specified according to the requirement.

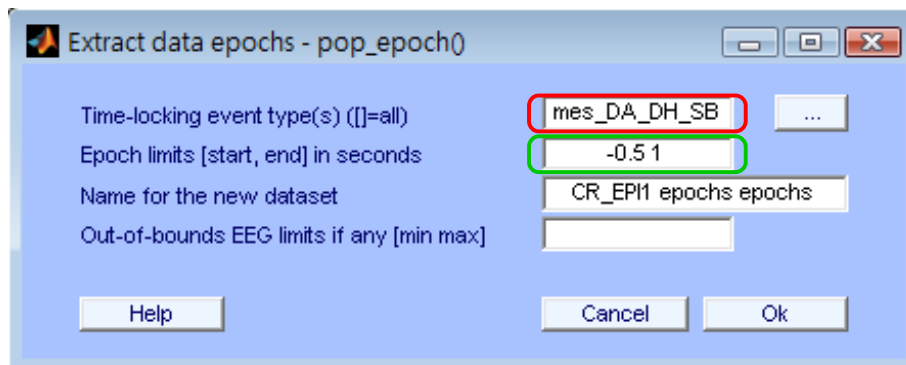


- Pop up Scroll channel activities window by: View→Continuous EEG

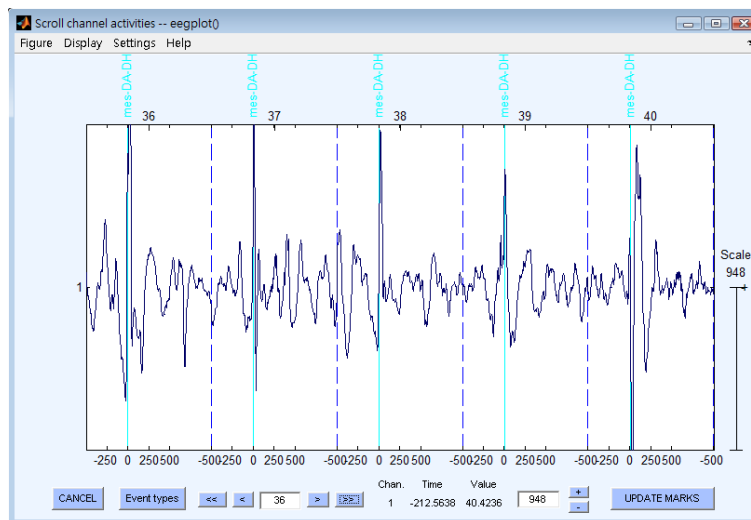


To epoch the dataset:

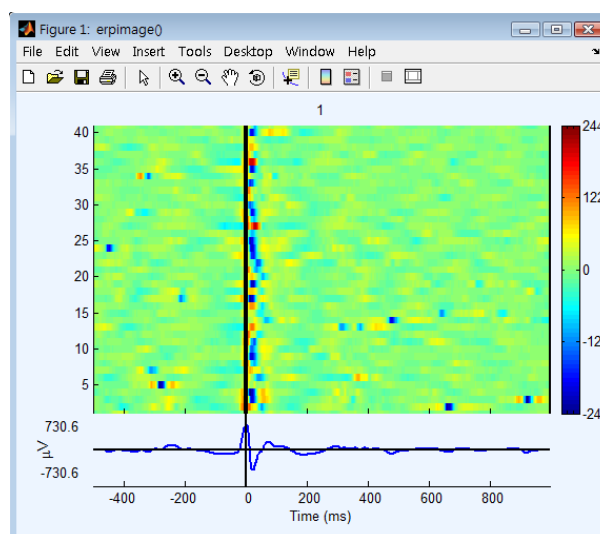
- Pop up by: Edit→Epoch
- Specify the **Time-locking event type(s)** (*the type of marker(s)*) and the **Epoch limits** (marked in red and green respectively), and click Ok.



3. Pop up Scroll channel activities window by: View→Continuous EEG

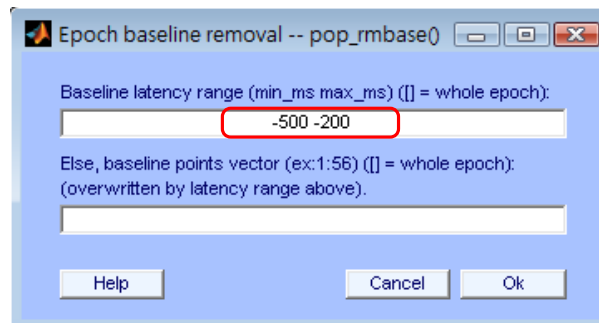


4. Pop up Channel ERP image window by: View→Epoched EEG

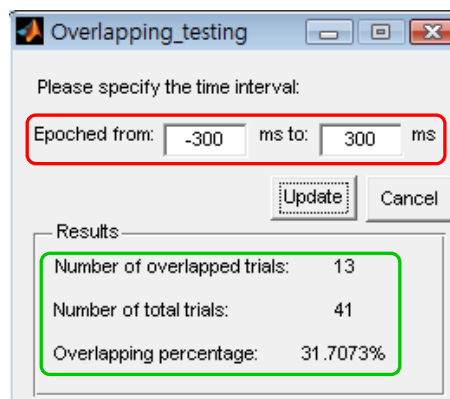


To perform Baseline correction (Optional, this step maybe not necessary in Epilepsy study):

1. Pop up by: Edit→Baseline correction
2. Specify the **Baseline latency range** and click Ok.



To check the number of overlapped trials (also the percentage), Click: View→Overlap testing. Specify the time interval that you want to check the overlapping (marked in red), and click Update. The testing results will be displayed in the “Results” panel (marked in green).



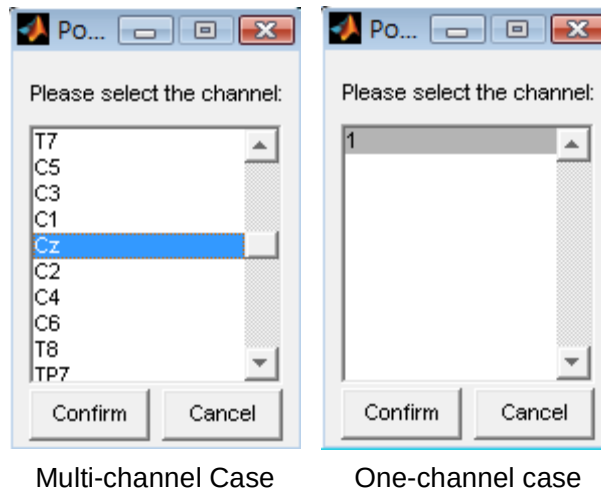
6 Single-trial analysis

6.1 Time domain

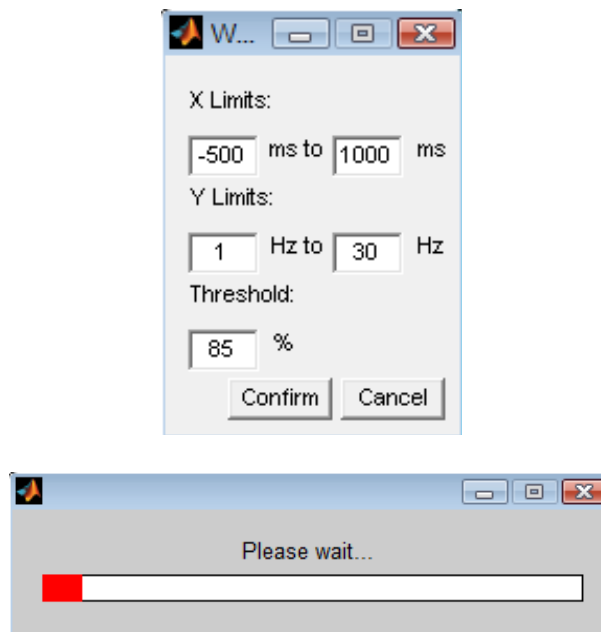
6.1.1 Wavelet Filtering (WF, filter only) *(not recommended in Epilepsy study)*

To increase the signal-to-noise ratio of single-trial EEG/ERP waveforms, Wavelet filtering, which is a type of time-frequency filtering, is recommended:

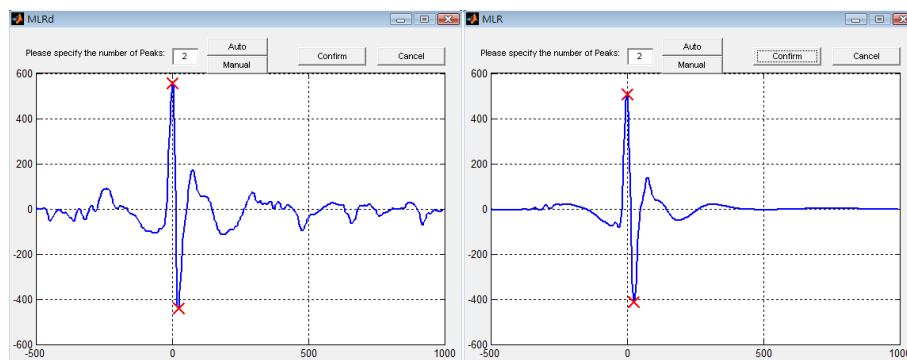
1. Pop up by: ST_A→Time domain→WF (filter only).
2. Select the Channel, which you are interested in, and click Confirm. Note that only one channel can be selected at one time.



3. Specify (1) X Limits (time); (2) Y Limits (frequency); and (3) Threshold value, and click Confirm. Note that this procedure will be **slow**, and a wait bar, which display the calculation progress, will be pop up.



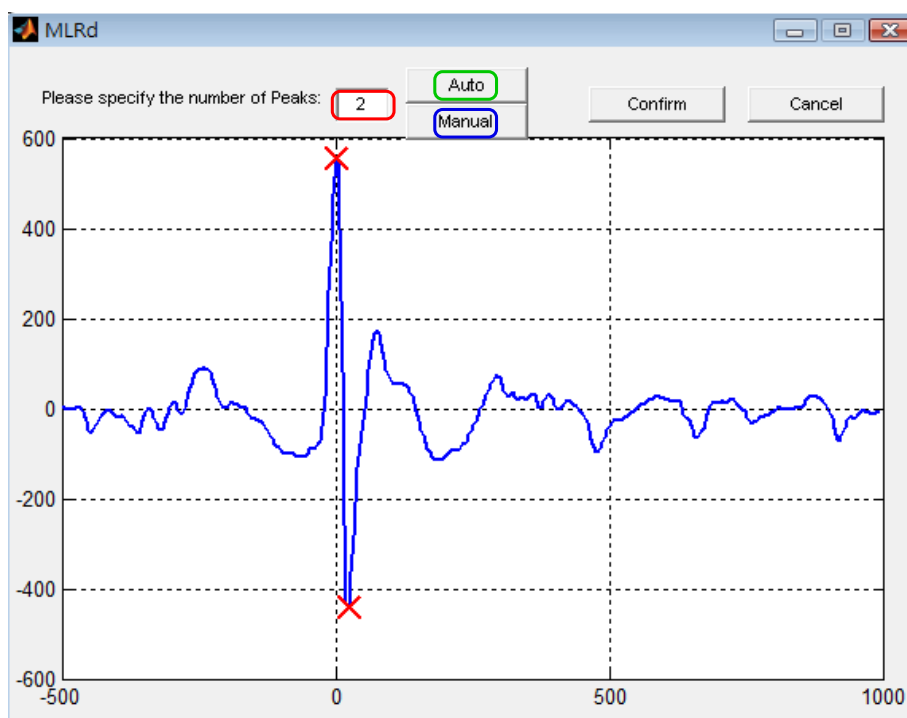
The following figure shows the comparison of waveform before (left panel) and after (right panel) wavelet filtering.



6.1.2 Multiple Linear Regression (MLR) *(not recommended in Epilepsy study)*

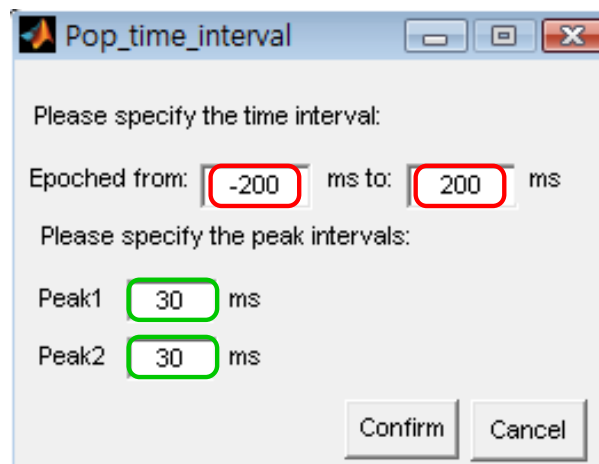
To perform time-domain single trial parameters detection using Multiple linear regression:

1. Pop up by: Edit→Time domain→MLR.
2. Select the Channel, which you are interested in, and click Confirm. Note that only one channel can be selected at one time.
3. Specify the **Number of peaks** and click either Auto or Manual to either automatically and manually select the peaks. The automatic selection will select the maximal/minimal peaks, which may or may not correct. In contrast, the manual selection will select the peaks that you are really interested in. Note that the peaks will be marked by red stars.



When you are satisfied with the selected peaks, click Confirm.

Specify the **Time interval** (the signal should be located at: marked in red) and the **peak intervals** (the peak should be located at: marked in green), and click confirm.



Pop_time_interval

Please specify the time interval:

Epoched from: ms to: ms

Please specify the peak intervals:

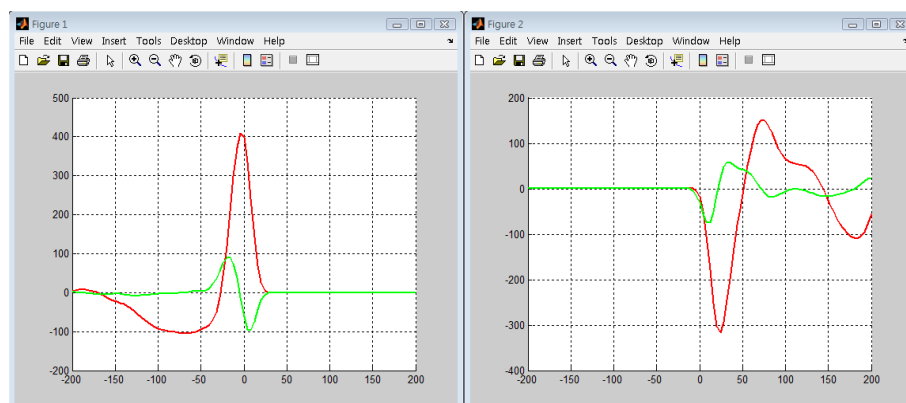
Peak1 ms

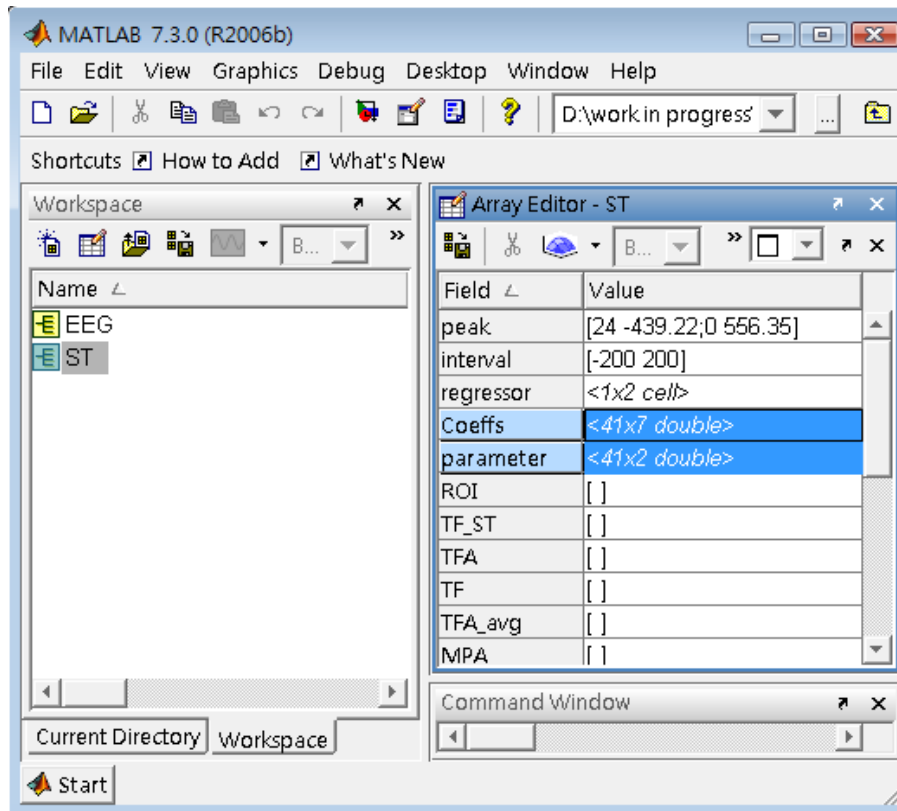
Peak2 ms

Confirm Cancel

Specify the **Time interval** (the signal should be located at: marked in red) and the **peak intervals** (the peak should be located at: marked in green), and click confirm.

When several plots, of which the number should be equal to the number of peaks, are displayed, the single-trial parameters (including peak amplitudes, latencies, and the corresponding coefficients of each regressor) have been estimated, and we can find the single-trial values at the Matlab workspace.

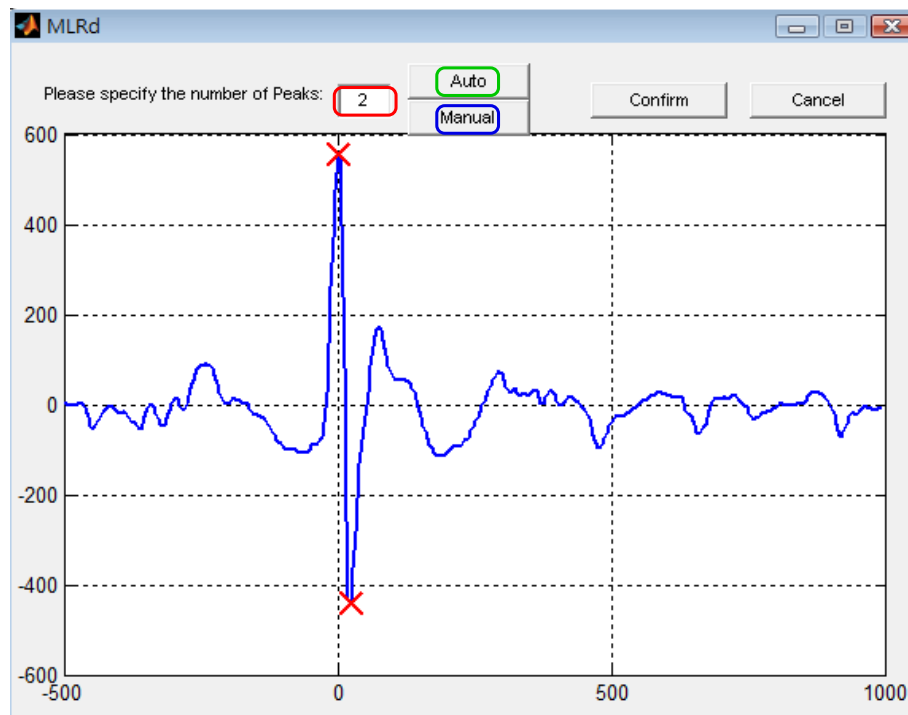




6.1.3 Multiple Linear Regression with Dispersion Term (MLRd)

To perform time-domain single trial parameters detection using Multiple linear regression with Dispersion term (MLRd) (Recommended in Epilepsy study):

1. Pop up by: Edit→Time domain→MLRd.
2. Select the Channel, which you are interested in, and click Confirm. Note that only one channel can be selected at one time.
3. Specify the **Number of peaks** and click either Auto or Manual to either automatically and manually select the peaks. The automatic selection will select the maximal/minimal peaks, which may or may not correct. In contrast, the manual selection will select the peaks that you are really interested in. Note that the peaks will be marked by red stars.



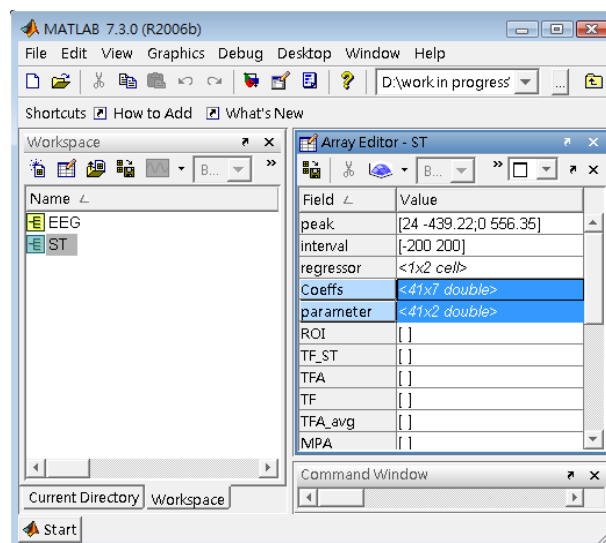
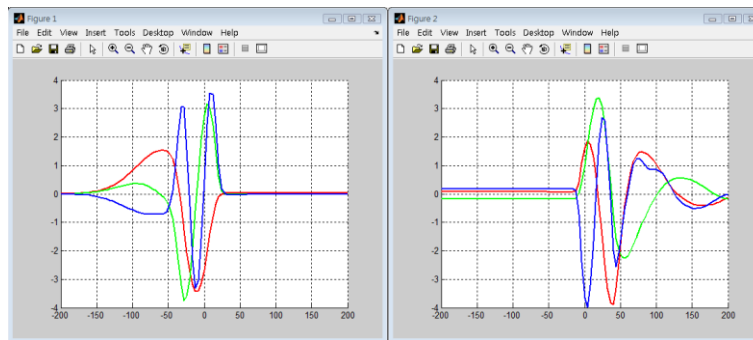
4. When you are satisfied with the selected peaks, click Confirm.
5. Specify the **Time interval** (the signal should be located at: marked in red) and the **peak intervals** (the peak should be located at: marked in green), and click confirm.

The figure shows a dialog box titled "Pop_time_interval". It contains the following fields and buttons:

- "Please specify the time interval:"
- "Epochd from:" followed by a text box containing "-200" (highlighted in red) and "ms to:" followed by a text box containing "200" (highlighted in red).
- "Please specify the peak intervals:"
- "Peak1" followed by a text box containing "30" (highlighted in green) and "ms".
- "Peak2" followed by a text box containing "30" (highlighted in green) and "ms".
- "Confirm" and "Cancel" buttons at the bottom right.

6. Specify the **Time interval** (the signal should be located at: marked in red) and the **peak intervals** (the peak should be located at: marked in green), and click confirm.
7. When several plots, of which the number should be equal to the number of peaks, are displayed, the single-trial parameters (including peak amplitudes, latencies, and the

corresponding coefficients of each regressor) have been estimated, and we can find the single-trial values at the Matlab workspace.



6.1.4 WF+MLR (not recommended in Epilepsy study)

The combination of wavelet filtering and multiple linear regression is to (1) increase the signal-to-noise ratio and (2) detect single-trial parameters accurately. The whole procedure can be start from: Edit→Time domain→WF+MLR. For a detail explanation of following procedures, please refer to **6.1.1 Wavelet Filtering** and **6.1.2 Multiple Linear Regression with Dispersion Term (MLRd)**.

6.1.5 WF+MLRd (not recommended in Epilepsy study)

The combination of wavelet filtering and multiple linear regression with dispersion term is to (1) increase the signal-to-noise ratio and (2) detect single-trial parameters accurately. The whole procedure can be start from: Edit→Time domain→WF+MLRd. For a detail explanation

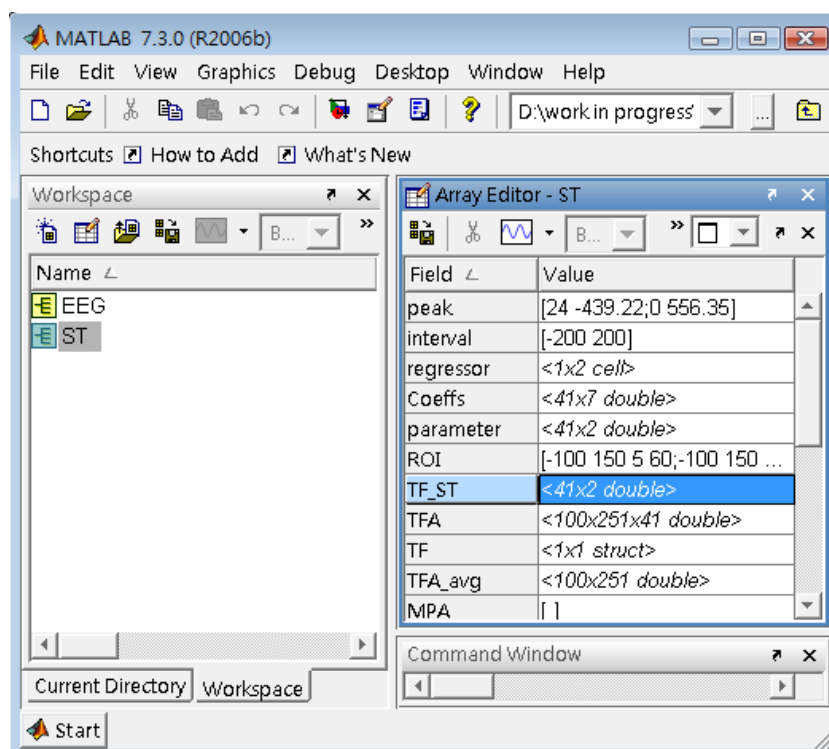
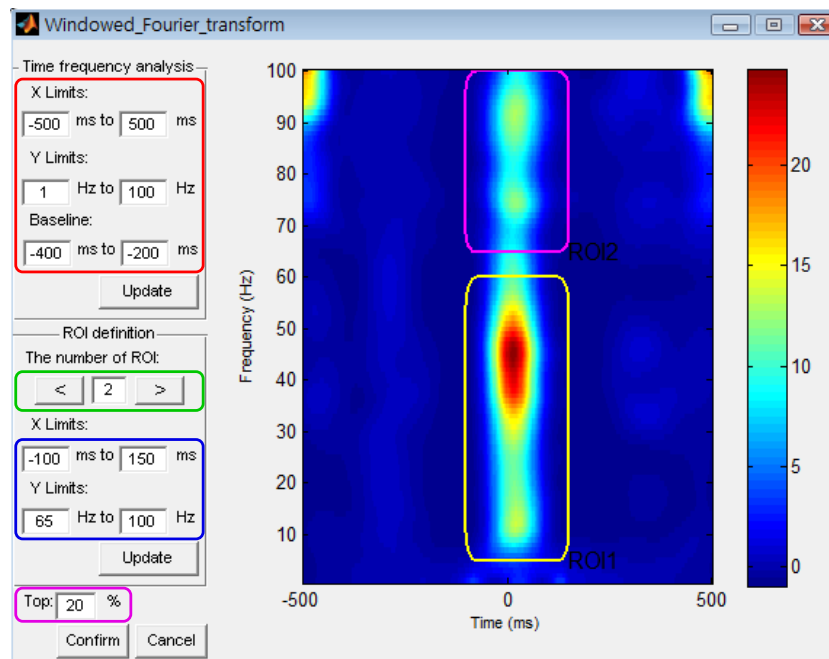
of following procedures, please refer to **6.1.1 Wavelet Filtering** and **6.1.3 Multiple Linear Regression (MLR)**.

6.2 Time-frequency domain

6.2.1 Windowed Fourier transform

To perform time-frequency-domain single trial parameters detection using Windowed Fourier transform (Recommended in Epilepsy study):

1. Pop up by: Edit→Time frequency domain→ Windowed Fourier transform.
2. Select the Channel, which you are interested in, and click Confirm. Note that only one channel can be selected at one time.
3. Specify (1) X Limits (time); (2) Y Limits (frequency); and (3) Baseline interval (time) (marked in red), and click Update. The baseline-corrected time-frequency representation will displayed at the right part of the current window.
4. Specify the current number of Region-of-interest (ROI) (marked in green). Note that this value should be start from 1, and can be sequentially changed once previous ROI(s) has/have defined.
5. Specify (1) X Limits (time) and (2) Y Limits (frequency) of each ROI (marked in Blue), and click Update. The area of each ROI will be marked using colorful rectangle.
6. Repeat step (4) and (5) until enough ROIs have been defined.
7. Specify the number of percentage (marked in Pink), where the top (in magnitude) percentage of voxels will be selected. Click confirm, and the single-trial parameters in each ROI (i.e. magnitudes) have been estimated, and we can find the single-trial values at the Matlab workspace.

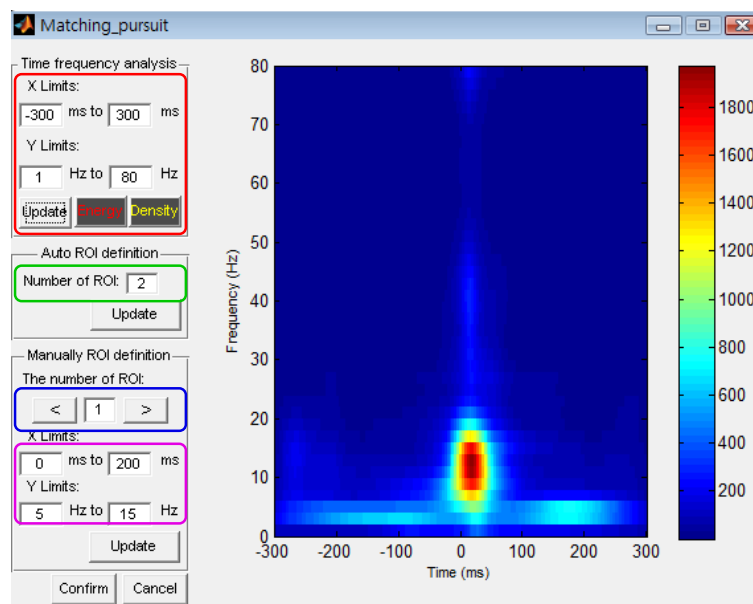
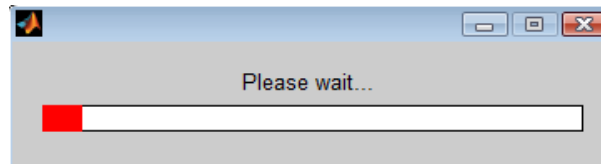


6.2.2 Matching Pursuit

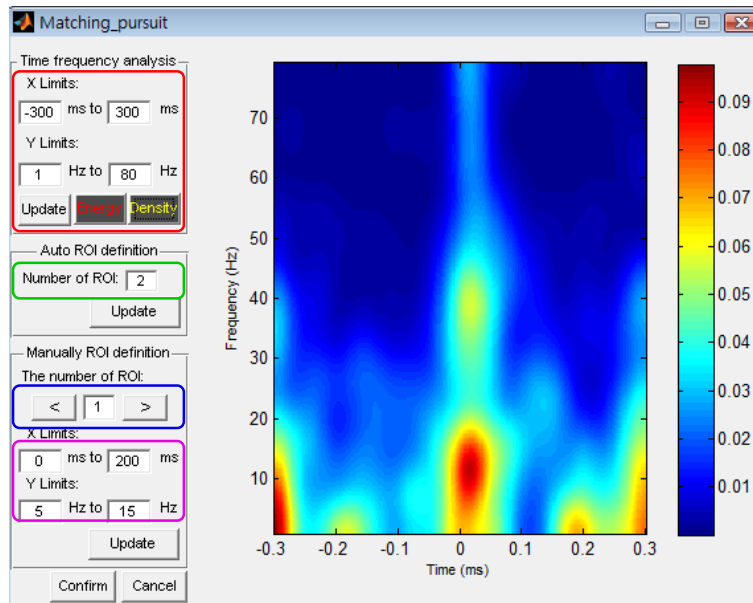
To perform time-frequency-domain single trial parameters detection using Matching Pursuit (Recommended in Epilepsy study):

1. Pop up by: Edit→Time frequency domain→ Matching Pursuit.

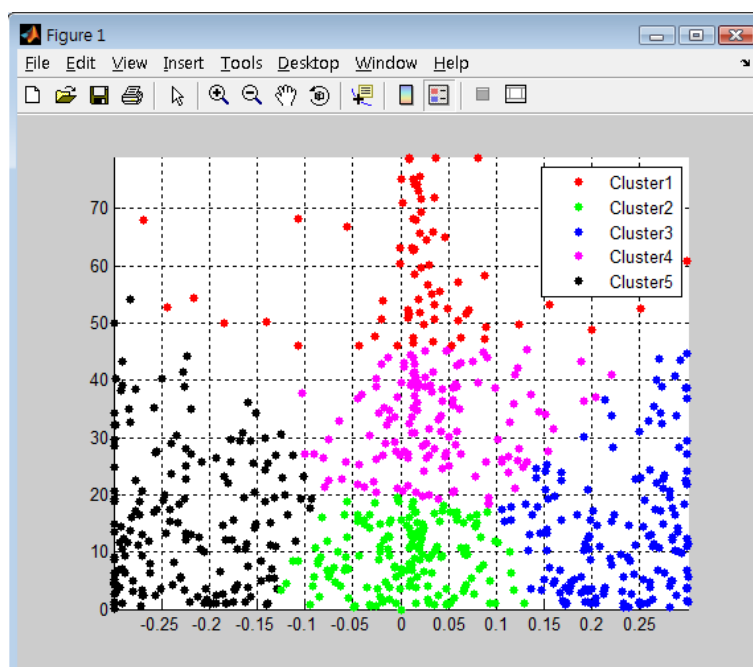
2. Select the Channel, which you are interested in, and click Confirm. Note that only one channel can be selected at one time.
3. Specify (1) X Limits (time) and (2) Y Limits (frequency) (marked in red), and click Update (**this will be very slow**). The wait bar will pop up, and display the calculation progress. Once finished, the time-frequency representation will displayed at the right part of the current window.



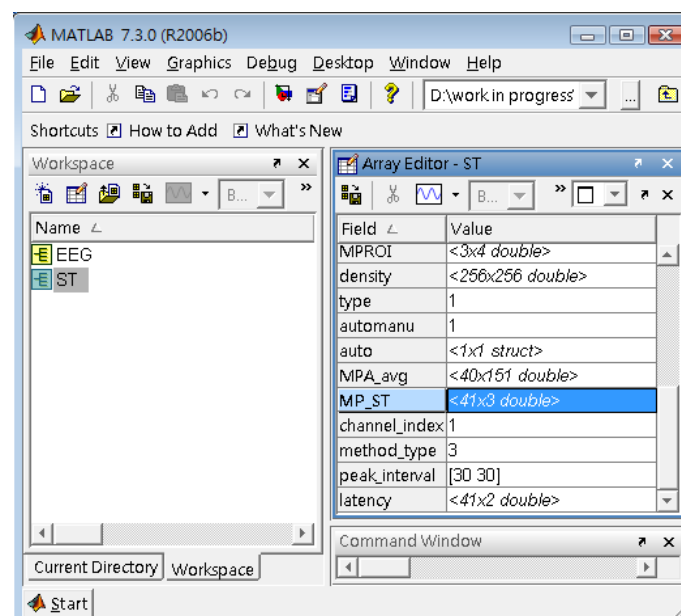
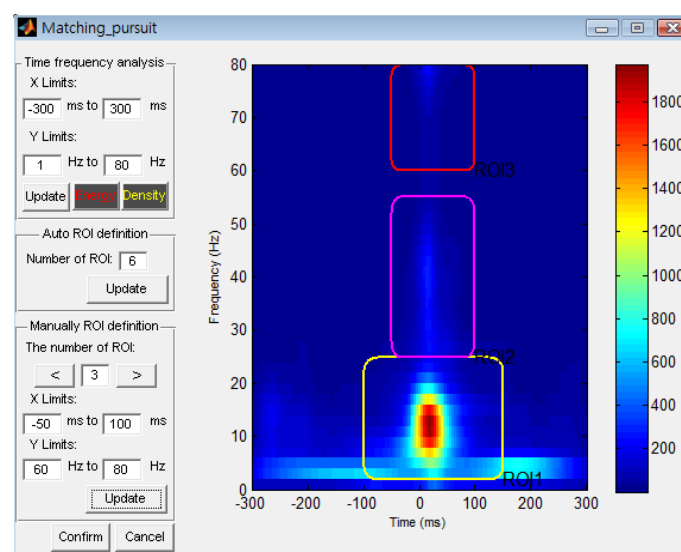
4. Click Energy or Density (marked in red), the time-frequency representation (represented as the Energy of the signal or as the density of the distribution of detailed time-frequency components) will displayed at the right part of the current window.



5. **(Automatic ROI definition)** The number of ROI can be defined using automatic approach (K-means clustering). Specify the number of ROI (marked in green), and click update. A plot with colored points, each color represents a cluster, will be pop up. If you are not satisfied with the clustering result, you can change the number of ROI, and click update again.



6. Specify the current number of Region-of-interest (ROI) (marked in blue). Note that this value should be start from 1, and can be sequentially changed once previous ROI(s) has/have defined.
7. Specify (1) X Limits (time) and (2) Y Limits (frequency) of each ROI (marked in pink), and click Update. The area of each ROI will be marked using colorful rectangle.
8. Repeat step (6) and (7) until enough ROIs have been defined.
9. Click confirm, and the single-trial parameters in each ROI (i.e. magnitudes) have been estimated, and we can find the single-trial values at the Matlab workspace.



How to define ROIs (my suggestions)?

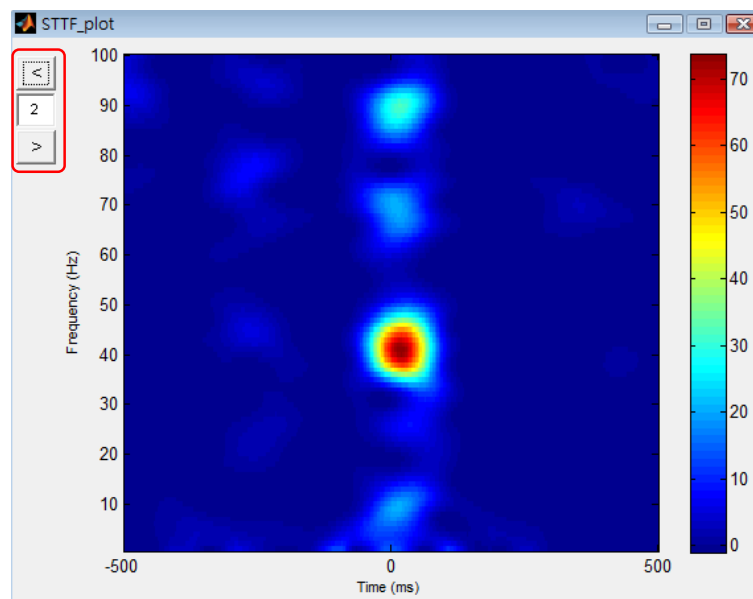
1. Based on previous publications.
2. Based on the WFT result (i.e. time-frequency distribution), MP results (i.e. time-frequency energy and time-frequency density), and K-means clustering result.
Merging all these information to decide a reasonable way to define the ROIs.
3. Based on the fMRI results.

7. Plotting Single-trial analysis results

To plot the generated regressors of all peaks, obtained by both MLR and MLRd, Click: View→Regressors.

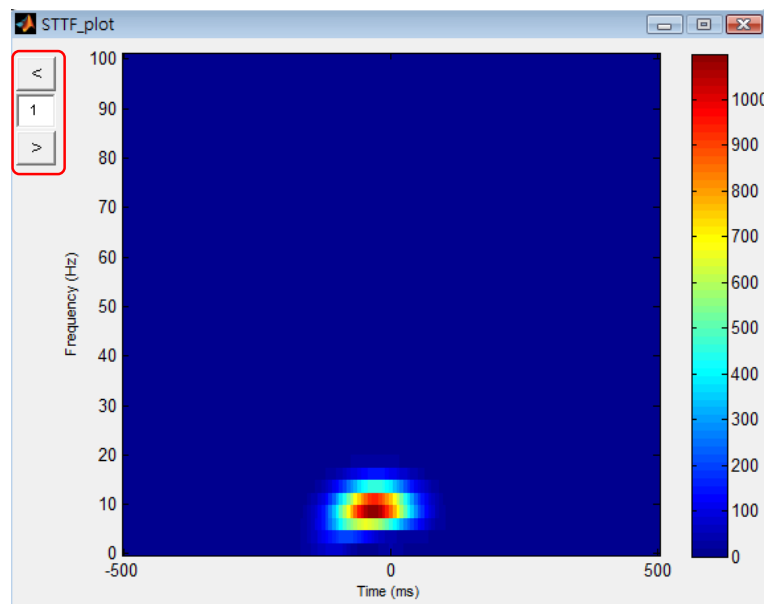
To plot the averaged time-frequency representation obtained using Windowed Fourier Transform, Click: View→Time frequency plot→Average(WFT).

To plot the single-trial time-frequency representation obtained using Windowed Fourier Transform, Click: View→Time frequency plot→Single trial(WFT). To scroll the time-frequency representation of each single trial, click “<” and “>” (marked in red).



To plot the averaged time-frequency representation obtained using Matching pursuit, Click:
View→Time frequency plot→Average(MP).

To plot the single-trial time-frequency detailed representation obtained using Matching pursuit, Click: View→Time frequency plot→Single trial(MP). To scroll the time-frequency representation of each single trial, click “<” and “>” (marked in red).



8. Saving estimated Single-trial values

To save the estimated single-trial values, both in the time domain and in the time-frequency domain:

1. Pop up save window by: File→Save→Save single trial values.
2. Specify the **file name and path**, and click Save.